DATA ITEM DESCRIPTION

Title: System Architecture Model (SAM)

Number: DI-SESS-82400
AMSC Number: F10366
DTIC Applicable: No
Preparing Activity: 19
Applicable Forms: N/A

Approval Date: 20221028
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2022-008

Use/relationship:

The purpose of the System Architecture Model (SAM) deliverable is to provide digital representation of the system architecture views that show and describe the Contractor's system architecture, invoking widely-accepted Model-Based Systems Engineering (MBSE) practices. The SAM serves as "the system architecture's authoritative source of truth" for all other System Development Life Cycle (SDLC) project deliverables (e.g., whenever there is a technical conflict between the SAM and another project deliverable, the SAM takes precedence).

This Data Item Description (DID) contains the format, content, and intended use information for the data deliverable resulting from the work tasks described in the contract.

Requirements:

1. **Reference Documents.** The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.
2. **Format.** Tool-Independent and Tool-Proprietary Formats. The SAM in its entirety, (i.e., including all profile changes, extensions, and modifications) shall be in all of the following formats: (a) the proprietary format of the Systems Modeling Language (SysML)-compliant modeling tool used to generate the SAM, (b) Extensible Markup Language (XML) Model Interchange (XMI), (c) HyperText Markup Language (HTML), and (d) Rich Text Format (RTF) or Portable Document Format (PDF).
3. **Content.** The SAM shall contain:

   3.1. Industry Standards Compliance and Conformance. The applicable industry standards are listed in prioritized order. Whenever there is a conflict among industry standards, the higher-priority industry standard shall prevail. The SAM shall be in conformance with the following industry standards:

   3.1.1. Object Management Group (OMG) Unified Architecture Framework (UAF). The SAM shall utilize the OMG UAF specification, version 1.0 or later (https://www.omg.org/spec/UAF). The SAM shall include UAF or Department of Defense Architecture Framework (DoDAF) views, identified by the Government.
   3.1.2. OMG SysML (https://www.omg.org/spec/SysML). The SAM shall be specified using the OMG SysML architecture modeling language, the de facto industry standard for graphically specifying systems engineering applications. The SAM shall be specified using

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OMG SysML, version 1.5 or later. The SAM shall follow OMG SysML well-formedness rules for notation (graphical syntax) and semantics.

3.1.3. *International Council on Systems Engineering (INCOSE) Systems Engineering Handbook* (https://www.incose.org/ProductsPublications/sehandbook) and *Guide to the Systems Engineering Body of Knowledge (SEBoK), Part 2: Foundations of Systems Engineering, Representing Systems with Models* (https://sebokwiki.org/wiki/Guide_to_the_Systems_Engineering_Body_of_Knowledge_(SEBoK).

3.1.4. International Organization for Standardization (ISO) / International Electrotechnical Commission (IEC) / Institute of Electrical and Electronics Engineers (IEEE) 15288:2015 – Systems and software engineering – System life cycle processes, section 6.4 Technical processes (https://www.ieee.org).

3.2. Architecture-Centric. The SAM shall define the overall System Context and the System Architecture of the System of Interest (SOI) throughout the SDLC. The System Context shall define the system scope in a clear and unambiguous manner for all SOI Stakeholders by explicitly defining the logical and physical interfaces external to the SOI. The System Architecture shall be defined with sufficient clarity and detail so that it will be self-describing and practical for a third party to bid and build the system without additional architecture support materials.

3.3. Enterprise Architecture Framework Conformance. The SAM shall conform to a UAF profile that organizes all SAM elements into SysML Views and SysML Packages, as appropriate.

3.4. Consistent Use of Conceptual Levels. All architectural relationships in the SAM shall maintain consistent use of conceptual levels. For example, all architectural elements defined in the SAM shall be at the same conceptual level (unless showing the internals of an element). Additionally, architecture elements shall be organized such that each element's conceptual level is clearly communicated.

3.5. Model-Based Simulation. The SAM shall provide model-based simulations of SysML behavioral diagrams (Activity, Sequence, and State Machine diagrams) and Parametric diagrams or it shall include integration points with MS&A tools, to include support for model-based simulations of SysML behavioral diagrams and model-based simulations of Parametric diagrams, as applicable.

3.6. Requirements Traceability and Impact Analysis. The SAM shall enable change impact analysis, whereby any change to any System Functional Requirement or Non-Functional Requirement shall elicit a list of SAM elements that are potentially affected (i.e., impacted or changed) by the subject Requirement change.

3.7. Model-Based Requirements Representation. All System functional requirements shall be defined as SysML Requirement elements in the SAM. All SysML Requirement elements shall be organized using appropriate SysML relationships (e.g., <<*Contains*>> and <<*DeriveReqt*>>). The SAM shall include a model-based requirements traceability matrix.

3.8. Use Cases and Intended System Purpose. All System Use Cases and the Intended System Purpose shall be defined as SysML Use Case elements. All SysML Use Case

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elements shall be organized using appropriate SysML relationships (e.g., <<*Include*>> and <<*Extend*>>).

3.9. System Structures. The SOI and each of its components shall be recursively specified as either black-box components and white-box components using SysML Block Definition Diagrams (BDDs) and SysML Internal Block Diagrams (IBDs), respectively. The recursive specification of the SOI composition structure shall start at the SOI Context level and descend into lower-level system elements until the Blocks being specified are at the lowest level needed by the current phase of the acquisition lifecycle. All system structures shall be organized using appropriate SysML relationships (e.g., *generalization*, *composition*, and *aggregation*). Each system structure shall be specified as a Block with defined Properties (e.g., Part Properties, Reference Properties, and Value Properties) and Proxy Ports, as appropriate. Proxy Ports shall be typed by Interface Blocks, and Interface Blocks shall be defined with Flow Properties that are typed by Signals.

3.10. Data Structures. Data structures shall be recursively specified using SysML BDDs. The recursive specification of data structures shall start with the highest-level data structures and descend into lower-level data structures, as needed by the current phase of the SDLC. All data structures shall be organized using appropriate SysML relationships (e.g., generalization, composition, and aggregation). Each data structure shall be defined by a Signal.

3.11. Dynamic Behavior. The dynamic behavior of the SOI and each of its components shall be specified by using an integrated combination of SysML behavioral diagrams (e.g., Sequence, Activity, State Machine diagrams). All System and Component dynamic behavior shall specify the following:

3.11.1. Information Flow and Item Flow Interface Interactions. Each SOI and component information flow and item flow interface interaction shall be included on at least one Sequence diagram or Activity diagram that shows both synchronous and asynchronous communication sequences and time constraints, as appropriate. Additionally, each information flow and item flow interaction shall appear in at least one associated IBD.

3.11.2. States and Modes. SOI states and modes shall be defined using state machine diagrams. The primary states and modes shall be specified. Each specified State and Mode defines at least one state/mode-specific behavior that identifies (behaviorally) why that state/mode is critical and needed to be identified.

3.12. Metamodel Relationships. The SAM shall contain a metamodel that includes the following relationships between model element types:

3.12.1. Static structural elements relate to requirement elements using the SysML <<satisfy>> relationship.

3.12.2. Behavioral activity elements relate to static structural elements using the SysML <<allocate>> relationship.

3.12.3. Behavioral use case elements relate to static structural elements using the SysML <<association>> relationship.

3.12.4. Requirement elements relate to document artifact elements using the SysML <<trace>> relationship.

3.12.5. Static structural elements relate to document artifact elements using the SysML <<trace>> relationship.

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3.12.6. Interface block elements relate to document artifact elements (e.g., Interface Control Document (ICD)) using the SysML <<*trace*>> relationship.
3.12.7. Static structural elements relate to other structural elements using the SysML <<allocate>> relationship.

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